Data Audit: Pre-Impact Footage in ACCIDENT

Purpose

The pre-impact footage in ACCIDENT is enough to train and evaluate a per-window onset classifier, so I will keep the cascade but make it temporal. Every clip in both splits contains an accident, so there are no normal clips to use as negatives or for counting false alarms. The only normal traffic is the part of each clip before the impact. This audit checks how much of that footage exists, especially clips with fewer than 8 frames before the accident.

Method

the file i created is `audit_metadata.py`

- fps = `no_frames / duration`, the same convention the ACCIDENT repo uses.
- Pre-impact time = `accident_time`; pre-impact frames = `accident_frame`.
- Post-impact time = `duration - accident_time`.
- Frames the model would see before the impact = `accident_time x model fps`, run at 8 fps and 4 fps.

Results

Almost every clip has usable footage before the accident: 0 synthetic clips and only 1 of my 400 test clips have fewer than 8 frames before the impact.

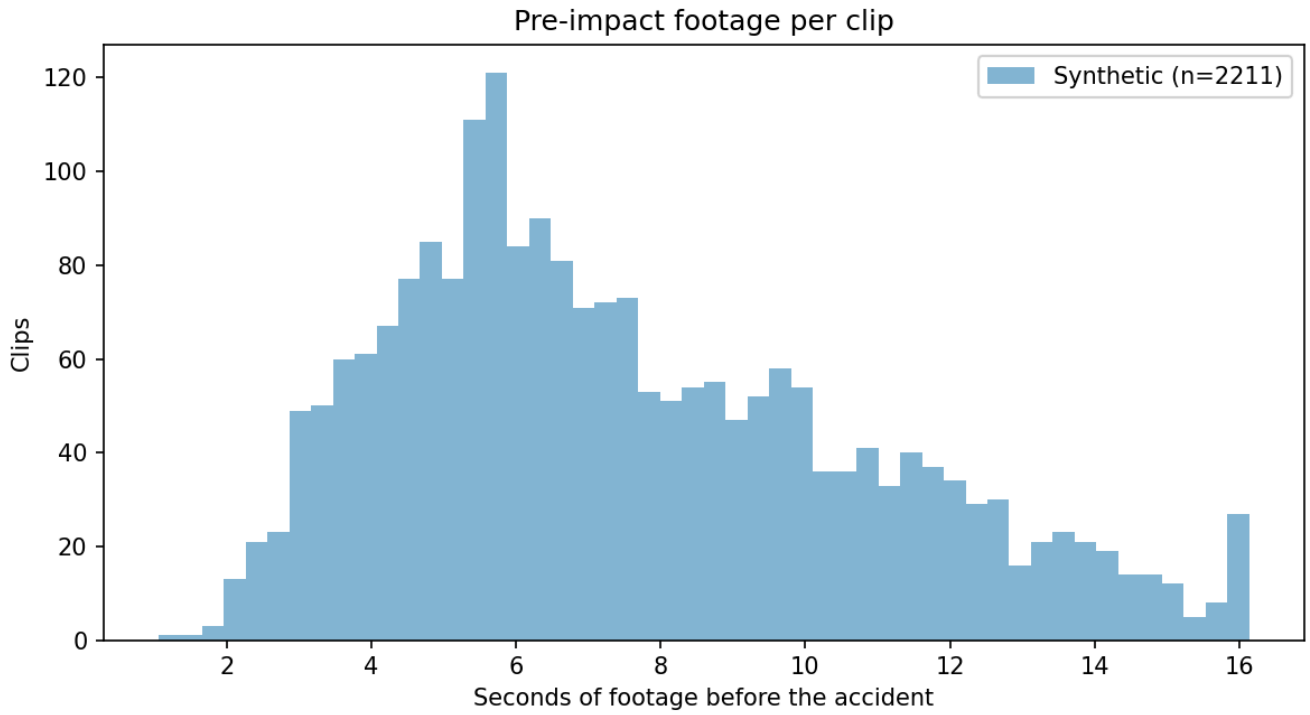
	Synthetic	Real (all)	Real (frozen test)
Clips	2,211	2,027	400
Frame rate	20 fps (all)	mostly 15 fps	mostly 15 fps
Duration, median (min-max)	17.6 s (5.8-32.2)	26.8 s (1.3-115.0)	28.3 s (1.3-115.0)
Pre-impact, median (min)	6.9 s (1.05)	4.7 s (0.12)	4.8 s (0.37)
Post-impact, median	9.9 s	19.7 s	19.2 s
Fewer than 8 frames before impact	0	8 (0.4%)	1 (0.2%)

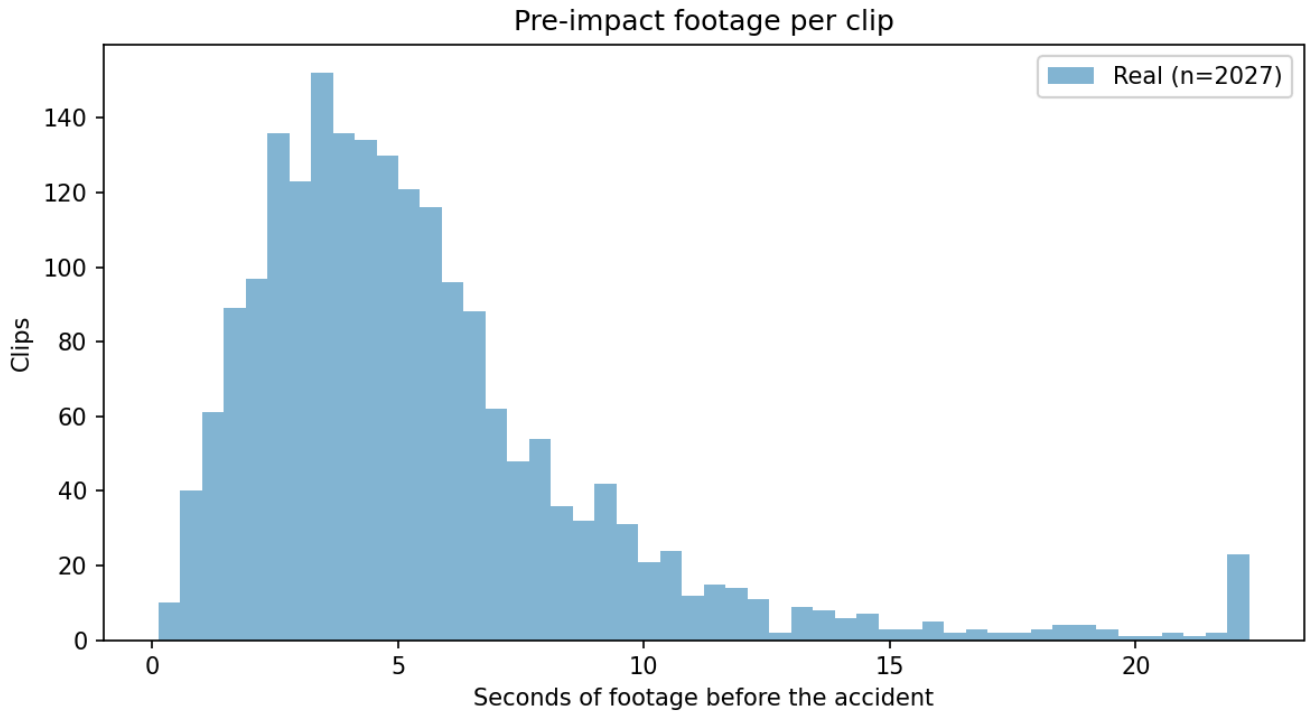
	Synthetic	Real (all)	Real (frozen test)
Under 1 s before impact	0	46 (2.3%)	10 (2.5%)
Under 2 s before impact	5 (0.2%)	215 (10.6%)	40 (10.0%)
Total pre-impact footage	4.67 h	3.13 h	0.66 h

Collision types differ a lot between the two domains:

Type	Synthetic	Real (all)	Real (frozen test)
rear-end	794 (35.9%)	328 (16.2%)	73 (18.2%)
head-on	588 (26.6%)	117 (5.8%)	23 (5.8%)
sideswipe	405 (18.3%)	245 (12.1%)	49 (12.2%)
t-bone	358 (16.2%)	657 (32.4%)	131 (32.8%)
single	66 (3.0%)	680 (33.5%)	124 (31.0%)

Sampling rate matters for synthetic clips. At 8 fps, only 5 clips (0.2%) cannot fit a full 16-frame window before the impact. At 4 fps that rises to 258 clips (11.7%).





Decisions

I will label each window by where it ends relative to the impact, and measure false alarms on the pre-impact part of the real test clips.

Labels. Labels use `accident_time` in seconds, not frame index, because about 16% of real clips have frame and time annotations that disagree (Week 2 fps audit).

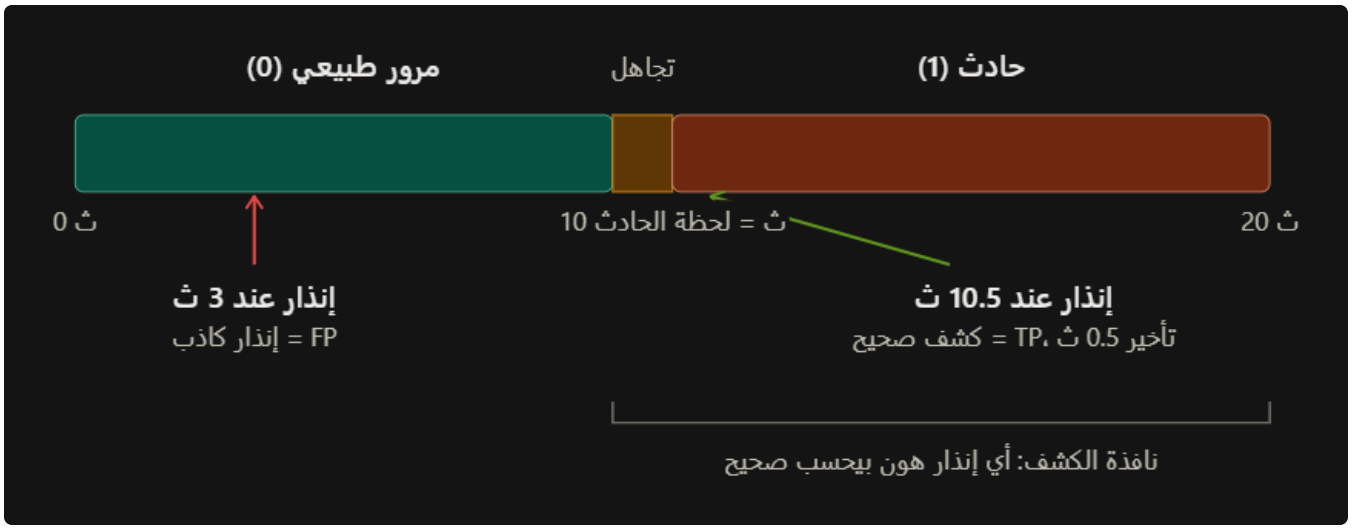
Window ends...	Label
more than 1 s before the impact	0 (normal)
in the last 1 s before the impact	ignored
at or after the impact	1 (accident)

Window. 16 frames sampled at 8 fps, so each window covers 2 s, with a 0.5 s stride. 8 fps keeps 99.8% of synthetic clips usable, and most real clips are 15 fps. Clips with too little pre-impact footage are padded with their first frame and reported separately (under 1 s, 1-2 s, over 2 s).

False alarms. An alarm more than 1 s before the impact is a false alarm (FP). No alarm between 1 s before and 10 s after the impact is a missed accident (FN). I will report:

- **Early-alarm rate** (main metric): share of test clips with at least one false alarm. Each clip is 0.25% of the 400.
- **False alarms per hour**: false alarms divided by about 0.55 h of usable pre-impact footage. One alarm moves this by about 1.8 per hour, so it is reported with that caveat.
- **Miss rate and detection delay** for the accidents themselves.

Thresholds are chosen on synthetic validation only.



Proposed changes

The main change is that the cascade decides per time window, not per clip, since every clip is an accident.

Proposal section	Change
6 Datasets	State that both splits contain only accidents and that normal data is the pre-impact footage.
7.1 Classifier	Per-window labels (0 / ignore / 1), 16 frames at 8 fps; class weighting for collision type, since single is 3% synthetic vs 33.5% real.
7.1 Ablation 1	Add random re-cropping of synthetic clips, because the impact comes earlier in real clips (4.7 s vs 6.9 s median), plus frame-rate variation.

Proposal section	Change
7.4 Cascade	The classifier scans the clip and proposes a window of about 2 s either side of the onset. Only uncertain clips go to the VLM, and only that window. Report accuracy against the share of clips sent to the VLM.
8 Evaluation	Keep the benchmark score on the 400 clips; add early-alarm rate, false alarms per hour, miss rate and delay. Break results down by quality, weather, day_time and pre-impact length.

questions and next steps:

1. should we add dataset for normal traffic only like UA-DETRAC ?

next steps Then I will build the window labelling pipeline.